

# **Business Analytics Case Study Report**

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### **Telco Customer Churn Analysis**

#### **Introduction**

Customer Churn is one of the major challenges faced by subscription-based businesses. It is the rate at which customers stop patronizing a company. This is a problem because the business starts to fail if there's a consistent decline in customer patronage. For a telecom company, high churn can lead to revenue loss, reduced customer base, and increased pressure to acquire new customers

This analysis aims to help the Telecom Company understand the reasons customers are leaving, identify trends, and recommend appropriate solutions. The goal of this analysis is to provide business insights and practical recommendations that can help improve customer retention.

#### **Business Problem Statement**

The telecom company is experiencing customer churn and needs to understand the factors contributing to customers leaving. This analysis aims to help the company identify customer segments with higher churn risk and recommend data-driven strategies to improve retention.

The key business question is:

**Why are customers leaving the telecom company, and what actions can the company take to reduce churn?**

## **Dataset Description**

The dataset used for this analysis is the Telco Customer Churn dataset. It contains customer-level information related to demographics, services subscribed to, contract type, payment method, charges, and churn status.

The dataset contains:

- 7,043 rows
- 21 original columns
- Customer profile fields such as gender, senior citizen status, partner, and dependents
- Service-related fields such as internet service, online security, tech support, streaming TV, and streaming movies
- Account-related fields such as tenure, contract type, payment method, monthly charges, and total charges
- Target column: Churn

The target column shows whether a customer churned or remained with the company.

## **Data Cleaning and Analysis Process**

The dataset was loaded and inspected using Python. The first step was to understand the structure of the data, check the number of rows and columns, inspect data types, and assess missing values.

The dataset had no duplicate records and no direct missing values. However, the TotalCharges column was originally stored as text instead of a numeric column. Further investigation showed that 11

records had blank values in TotalCharges, and all of those customers had tenure = 0. This suggested that they were likely new customers who had not yet accumulated total charges. These blank values were replaced with 0, and the column was converted to a numeric data type.

Additional columns were created to support the analysis:

- **ChurnFlag:** numeric version of churn, where 1 represents churned customers, and 0 represents retained customers.
- **TenureGroup:** grouped customers based on tenure duration.
- **MonthlyChargeGroup:** grouped customers based on monthly charge levels.
- **Churn Status:** used in Power BI to separate churned and retained customers.

The analysis focused on identifying customer groups with churn rates higher than the overall churn rate.

## Key Findings

The overall churn rate was 26.54%, meaning that about 27 out of every 100 customers left the company.

## Contract Type

Customers on month-to-month contracts had the highest churn rate at 42.71%. One-year contract customers had a churn rate of 11.27%, while two-year contract customers had the lowest churn rate at 2.83%.

This shows that customers on shorter contracts are more likely to leave.

## Tenure

Customers with shorter tenure had higher churn rates. Customers within the 0–12 months tenure group had the highest churn rate at 47.44%. Customers in the 49–72 months tenure group had the lowest churn rate at 9.51%.

This suggests that newer customers are more likely to churn, while long-term customers are more loyal.

## **Monthly Charges**

Customers in the high monthly charge group had the highest churn rate at 35.36%. Medium-charge customers had a churn rate of 23.94%, while low-charge customers had the lowest churn rate at 10.89%.

This suggests that customers paying higher monthly fees may be more sensitive to pricing or may expect greater value from the service.

## **Internet Service**

Fiber optic customers had the highest churn rate among internet service users at 41.89%. DSL customers had a churn rate of 18.96%, while customers with no internet service had the lowest churn rate at 7.40%.

Since fiber optic customers also had higher average monthly charges, their churn may be connected to pricing, perceived value, or service expectations.

## **Payment Method**

Customers using electronic checks had the highest churn rate at 45.29%. Customers using mailed checks had a churn rate of 19.11%, while customers using automatic bank transfer and credit card had lower churn rates of 16.71% and 15.24%, respectively.

This suggests that customers using automatic payment methods may be more stable, while electronic check users represent a higher-risk customer group.

## **Business Insights**

The analysis shows that customer churn is strongly linked to contract type, tenure, monthly charges, internet service type, and payment method.

Customers on month-to-month contracts are at higher risk because they are less tied to the company. This makes them easier to lose compared to customers on one-year or two-year contracts.

New customers are also a high-risk segment. The first 12 months appear to be a critical period for customer retention. If the company fails to build trust and satisfaction early, customers may leave before becoming loyal.

High-paying customers also show higher churn. This means the company may be losing valuable customers who contribute more revenue. These customers may expect better service quality, stronger support, or more value for the price they pay.

Fiber optic customers also require attention because they have the highest churn among internet service users. While the dataset does not directly measure service quality, the higher churn rate among fiber optic users suggests that the company should monitor their satisfaction, complaints, and service experience closely.

The payment method also provides useful insight. Customers using electronic checks are more likely to churn, while customers using automatic payment methods are less likely to leave. This may indicate that automatic payment users are more stable or more committed.

## **Recommendations**

### **1. Encourage longer-term contracts**

The company should create incentives that encourage month-to-month customers to move to one-year or two-year contracts. This can include discounts, loyalty rewards, service bundles, or promotional offers. Since customers on longer contracts have much lower churn rates, this strategy can improve customer retention.

### **2. Strengthen first-year customer retention**

The company should focus more on customers within their first 12 months because they have the highest churn rate. New customers can be supported through better onboarding, welcome offers, regular check-ins, quick complaint resolution, and personalized retention messages.

### **3. Review pricing and improve value for high-paying customers**

Since customers with higher monthly charges churn more, the company should review its pricing structure and make sure customers feel they are getting good value for the amount they pay. This can include loyalty discounts, flexible pricing plans, better bundles, or special offers for high-value customers.

### **4. Improve fiber optic customer satisfaction**

Fiber optic customers have the highest churn among internet service users and also pay the highest average monthly charges. The company should ensure that fiber optic customers receive reliable service, strong technical support, and enough value to

justify the higher cost. Although the dataset does not directly measure service quality, monitoring complaints and service performance would help the company understand this group better.

## **5. Encourage automatic payment methods**

The company should educate customers on the benefits of automatic payment methods and encourage them to switch from electronic checks to automatic bank transfers or credit card payments. This can be supported through simple setup processes, reminders, small discounts, or billing incentives.

## **Conclusion**

This analysis identified several important factors associated with customer churn in the telecom company. The highest churn risks were found among month-to-month customers, new customers, high-charge customers, fiber optic users, and electronic check users.

The overall churn rate of 26.54% shows that customer retention should be a priority for the company. By encouraging longer-term contracts, improving first-year customer support, reviewing pricing value, improving fiber optic customer experience, and promoting automatic payment methods, the company can reduce churn and strengthen customer loyalty.

The findings from this analysis can support business managers in making data-driven decisions that improve customer retention and long-term business growth.